

Michael Mitry

AI & Full-Stack Engineer



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Multidisciplinary Engineer with a unique background encompassing Architecture, Commercial Aviation, and Filmmaking, who successfully transitioned into the AI and software development world. I bridge the gap between structural design principles and highly functional, modern web applications, and leverage generative AI to create advanced architectural and cinematic visualizations. Specializing in the React ecosystem (React 19, TypeScript, Tailwind CSS) and serverless architectures (Firebase, Google Cloud), I build complex systems rapidly by leveraging AI-assisted "vibe coding" workflows. I bring deep, hands-on domain expertise to aviation and simulation software, translating intricate business and regulatory requirements into scalable, elegant solutions. I am constantly learning and adopting new tools to stay ahead of the rapidly accelerating pace of the tech market.

📁 PROFESSIONAL EXPERIENCE

03/2024 – Present

Filmmaker & Visual Effects Artist, Society Of PLAY

- Create high-end visual effects to enhance storytelling and production quality, requiring deep technical understanding of rendering pipelines and timeline management.
- Mentor junior editors, providing guidance on complex software techniques and visual effects workflows.

06/2023 – 08/2023

Architectural Trainee, ProfilerZ Consultant & Hill International

- Constructed complex architectural models from abstract ideas, optimizing efficiency and quality in the architectural visualization process.
- Developed systematic approaches to handle multiple concurrent tasks and ensure smooth project workflows.

🎓 EDUCATION & CERTIFICATIONS

02/2024
Egypt

Bachelor of Architecture and Environmental Design, Arab Academy For Science And Technology

- Graduation Project: "ReviveTech" (Grade: A-)
- Achievements: Student Activities Media Head (Managed university media content production and software platforms)



TECHNICAL SKILLS

Frontend Development — React (v19), TypeScript, Tailwind CSS v4, shadcn/ui, Framer Motion, HTML5/CSS3 • **Backend & Cloud** — Node.js, Express, Firebase (Firestore, Auth, Admin SDK), Google Cloud Run, Docker • **Tooling & Libraries** — Vite, esbuild, Git, pdf-lib (Document generation), papaparse (Data processing), recharts (Data visualization) • **UI/UX & Web Design** — Figma, Advanced Web Design • **Architecture & AI Visualization** — Generative AI for Architecture, Revit, AutoCAD, Rhinoceros, Grasshopper, SketchUp, Blender, Cinema4D • **Filmmaking & Video Production** — Cinematography, Video Editing, Premiere Pro, DaVinci Resolve, Vegas Pro, After Effects, FL-Studio • **Other Tools** — Excel



SELECTED SOFTWARE ENGINEERING LABS & PROJECTS

2024 – Present

Strike Boxing CRM, Lead Developer

- Architected and developed a full-fledged, custom Customer Relationship Management (CRM) system for a fitness/boxing club from scratch.
- Frontend: Built a responsive, highly interactive UI using React 19, TypeScript, and Tailwind CSS v4. Implemented complex data visualization using Recharts.
- Backend: Engineered a serverless backend using Firebase (Firestore and Auth) with Node.js/Express, deployed on Google Cloud Run via Docker.
- Features: Implemented advanced sales attribution logic, complex audit logging with historical lookups, automated PDF reporting, CSV data parsing, and QR code integration.

2024 – 2024

ATPL Vector (atplvector.com), Frontend Developer

- Developed a comprehensive, interactive e-learning platform for Airline Transport Pilot Licence (ATPL) students.
- Engineered complex interactive components, including an SVG aircraft inspector, PANS-OPS holding calculators, and regulatory compliance visualizations.
- Translated dense EASA regulations (Part-FCL, Part-MED, Rules of the Air) into dynamic architectural trees, comparison tables, and fully searchable glossaries using modern UI paradigms.



EDUCATION & CERTIFICATIONS

Commercial Pilot

- Licensed Commercial Pilot.
- Deep practical and theoretical knowledge of EASA regulations, IFR procedures, and PANS-OPS.